

29.10.2024

LRT:  $\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} \mathcal{L}(\theta|x)}{\sup_{\theta \in \Theta} \mathcal{L}(\theta|x)}$  If  $H_0: \theta \in \Theta_0$  vs.  $H_1: \theta \in \Theta_1$  Para Inf  
 If  $\Theta_0 \cup \Theta_1 \subset \Theta$

$$\downarrow$$

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} \mathcal{L}(\theta|x)}{\sup_{\theta \in \Theta_0 \cup \Theta_1} \mathcal{L}(\theta|x)}$$

(i)  $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$   $H_0: \mu = \mu_0$  vs.  $H_1: \mu \neq \mu_0$

(i) If  $\sigma$  known: LRT:  $\phi(x) = \mathbb{1}\left(\frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma} > z_{\alpha/2}\right)$

→ UMPU Test as well

(ii) If  $\sigma$  unknown: LRT:  $\phi(x) = \mathbb{1}\left(\frac{\sqrt{n}|\bar{x} - \mu_0|}{s} > t_{n-1, \alpha/2}\right)$

→ UMPU Test

Ex.  $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ . Derive LRT for  $H_0: \mu \leq \mu_0$  vs.  $H_1: \mu > \mu_0$   
 Likelihood → unknown

$$\mathcal{L}(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2} = (n-1)s^2 + n(\bar{x} - \mu)^2$$

Under  $H_0: \mu \leq \mu_0$  Fix  $\sigma > 0$

$$\max_{\mu} \text{ w.r.t. } \mu \iff \min_{\mu} (x - \mu)^2$$

The MLE of  $\mu$ ,  $\hat{\mu}_0 = \begin{cases} \bar{x} & \text{if } \bar{x} \leq \mu_0 \\ \mu_0 & \text{if } \bar{x} > \mu_0 \end{cases}$

$$\underset{\sigma}{\text{maximize}} (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum (x_i - \hat{\mu}_0)^2}$$

MLE of  $\sigma^2$  is  $\hat{\sigma}_0^2 = \frac{1}{n} \sum (x_i - \hat{\mu}_0)^2$

$$\sup_{\mu \leq \mu_0, \sigma^2} \mathcal{L}(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} e^{-n/2}$$

Under  $H_0 \cup H_1$

$$\text{MLE: } \hat{\mu} = \bar{x}$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$$

$$\sup_{\mu, \sigma^2} \mathcal{L}(\mu, \sigma^2) = (2\pi \hat{\sigma}^2)^{-n/2} e^{-n/2}$$

$$\lambda(x) = \left( \frac{\hat{\sigma}_0^2}{\hat{\sigma}^2} \right)^{-n/2} = \left( \frac{\sum (x_i - \hat{\mu}_0)^2}{\sum (x_i - \bar{x})^2} \right)^{-n/2}$$

$$= \left( \frac{\sum (x_i - \bar{x} + \bar{x} - \hat{\mu}_0)^2}{\sum (x_i - \bar{x})^2} \right)^{-n/2}$$

$$= \left( 1 + \frac{n(\bar{x} - \hat{\mu}_0)^2}{(n-1)\hat{\sigma}^2} \right)^{-n/2} < K$$

$$\lambda(x) < K \Leftrightarrow \frac{\sqrt{n}|\bar{x} - \hat{\mu}_0|}{S} > c^* \quad (c^* > 0)$$

$$\mathbb{P}_{\mu_0}[\lambda(x) < K] = \mathbb{P}_{\mu_0} \left[ \underbrace{\frac{\sqrt{n}|\bar{X} - \hat{\mu}_0|}{S} > c^*, \bar{X} \leq \mu_0}_{\hat{\mu}_0 = \bar{X}} + \underbrace{\frac{\sqrt{n}|\bar{X} - \hat{\mu}_0|}{S} > c^*, \bar{X} > \mu_0}_{\hat{\mu}_0 = \mu_0} \right]$$

$$= \mathbb{P}_{\mu_0} \left[ \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > c^* \right] = \alpha$$

$\sim t_{n-1}$

$$c^* = t_{n-1, \alpha}$$

LRT:  $\phi(x) = \mathbb{1} \left( \frac{\sqrt{n}(\bar{x} - \mu_0)}{S} > t_{n-1, \alpha} \right)$

(To show that  $\mathbb{P}_{\mu} \left( \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > c^* \right) \leq \alpha$   $\forall \mu \leq \mu_0$ )

Ex.  $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu_0, \sigma^2)$   
known

Derive LRT for  $H_0: \sigma = \sigma_0$  vs.  $H_1: \sigma \neq \sigma_0$

$$\mathcal{L}(\sigma^2) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2} \frac{\sum (x_i - \mu_0)^2}{\sigma^2}} = Q$$

Under  $H_0$   $\mathcal{L}(\sigma_0^2) = \sup_{H_0} \mathcal{L}(\sigma^2) = (2\pi\sigma_0^2)^{-n/2} e^{-\frac{1}{2} \frac{Q}{\sigma_0^2}}$

Under  $H_0 \cup H_1$   $\sup_{\sigma} \mathcal{L}(\sigma^2) = (2\pi \frac{Q}{n})^{-n/2} e^{-n/2}$

LRT Stat  $1(x) = \left( \frac{\sigma_0^2}{Q/n} \right)^{-n/2} e^{-1/2 (Q/\sigma_0^2 - n)} < K$

$$\Leftrightarrow \left( \frac{Q}{\sigma_0^2} \right)^{n/2} e^{-1/2 Q/\sigma_0^2} < K^*$$

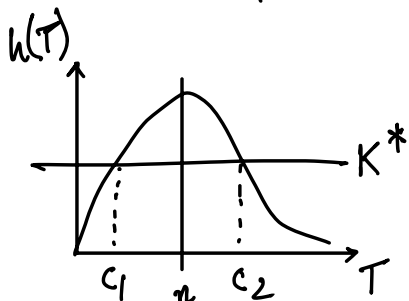
$$T = \frac{Q}{\sigma_0^2} \underset{H_0}{\sim} \chi_n^2$$

$$\Leftrightarrow \underbrace{T^{n/2} e^{-T/2}}_{h(T)} < K^*$$

$$\frac{\partial}{\partial T} \ln h(T) = \frac{\partial}{\partial T} \left( \frac{n}{2} \ln T - \frac{T}{2} \right)$$

$$= \frac{n}{2T} - \frac{1}{2} \longrightarrow T = n$$

$$\frac{\partial^2}{\partial T^2} \ln h(T) < 0$$



$$h(c_1) = h(c_2) = K^*$$

Reject  $H_0$  if  $T < c_1$  or  $T > c_2$

$$\text{where } c_1 \text{ and } c_2 \text{ s.t. } \begin{cases} P_{\sigma_0}(T < c_1) + P_{\sigma_0}(T > c_2) \\ h(c_1) = h(c_2) \end{cases} = \alpha$$

## UMPU Test

Reject  $H_0$  if  $T < K_1$  or  $T > K_2$

where  $K_1, K_2: P_{\sigma_0}(T < K_1) + P_{\sigma_0}(T > K_2) = \alpha$

$$E_{\sigma_0}(T \psi(T)) = \alpha E_{\sigma_0} T = \alpha n$$

$$\int_{-\infty}^{K_1} t f_T(t) dt + \int_{K_2}^{\infty} t f_T(t) dt$$

Ex  $X_1, \dots, X_n \sim N(\mu, \sigma^2)$

$$\mu = \mu_0 \text{ vs. } \mu > \mu_0$$

## Two-sample

$$\begin{cases} X_1, \dots, X_{n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2) \\ Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2) \end{cases}; \sigma_1^2, \sigma_2^2 \text{ known}$$

H.W.  $H_0: \mu_1 = \mu_2$  vs.  $H_1: \mu_1 \neq \mu_2$

$$\phi(\underline{x}, \underline{y}) = \mathbb{1} \left( \frac{|\bar{x} - \bar{y}|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} > \bar{z}_{\alpha/2} \right)$$

## Two-sample t-test

$$\begin{cases} X_1, \dots, X_{n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma^2) \\ Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma^2) \end{cases}$$

$$\text{LRT: } \phi(\underline{x}, \underline{y}) = \mathbb{1} \left( \frac{|\bar{x} - \bar{y}|}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} > t_{n_1+n_2-2, \alpha/2} \right)$$

If  $\sigma_1^2 \neq \sigma_2^2$  unknown: Satterwaite-Welch Test

$$\phi = \mathbb{1} \left( \frac{|\bar{x} - \bar{y}|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} > t_{\hat{\nu}}, \alpha/2 \right)$$

Satterwaite approx

$$\frac{\bar{x} - \bar{y}}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \stackrel{d}{\approx} t_{\hat{\nu}}$$

$$\text{where } \hat{\sigma}^2 = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{\frac{S_1^4}{n_1^2(n_1-1)} + \frac{S_2^4}{n_2^2(n_2-1)}}$$

Two-sample

$$\left[ \begin{array}{l} X_1, \dots, X_{n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2) \\ Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2) \end{array} \right] \text{ indep } H_0: \sigma_1^2 = \sigma_2^2 \text{ vs. } H_1: \sigma_1^2 \neq \sigma_2^2$$

$$\text{LRT: } \frac{S_1^2}{S_2^2} \rightarrow \frac{1}{n_1-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \rightarrow \frac{1}{n_2-1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

$$\frac{(n_i-1)S_i^2}{\sigma_i^2} \sim \chi_{n_i-1}^2 \quad * i=1,2$$

$$X = \frac{S_1^2}{S_2^2} \xrightarrow{H_0} \frac{(n_1-1)S_1^2/\sigma_0^2(n_1-1)}{(n_2-1)S_2^2/\sigma_0^2(n_2-1)} \sim F_{n_1-1, n_2-1}$$

LRT rej  $H_0$  if

$$\frac{S_1^2}{S_2^2} > F_{n_1-1, n_2-1, \alpha_1} \text{ or } < F_{n_1-1, n_2-1, 1-\alpha_2}$$

need another condition to find  $\alpha_1, \alpha_2$

## Confidence Set/Interval

$X_1, \dots, X_n \sim f(x|\theta) \quad \theta \in \Theta$ . Estimate  $\theta$

Already have point estimate =  $\hat{\theta}$

How confident are we that  $\hat{\theta} = \theta$ ?

1. No confidence (if  $\hat{\theta}$  is cont r.v.)

i.e.  $P_{\theta}(\hat{\theta}(X) = \theta) = 0$  if  $\hat{\theta}(X)$  cont. r.v.

Instead of point estimate  $\hat{\theta}$ , we would like to give a lower bound  $L(X)$  and an upper bound  $U(X)$  s.t.

$$\mathbb{P}_{\theta} (L(X) \leq \theta \leq U(X)) \geq \underbrace{1-\alpha}_{\text{confidence level}} \quad \theta \in \mathbb{R}$$

If  $\theta \in \mathbb{R}$  can have  $L(X) = -\infty$  or  $U(X) = \infty$

(1)  $[L(X), \infty)$  = lower confidence interval

(2)  $(-\infty, U(X)]$  = upper " "

(3)  $[L(X), U(X)]$  = two-sided " "

Motivating Ex.: (1)  $X$  = time to failure of an equipment  $\mu = \mathbb{E}X$

We want  $L(X)$   $\mathbb{P}(\mu \geq L(X)) \geq 1-\alpha$

(2)  $X$  = toxicity of a drug (adverse side effect)  $\mu = \mathbb{E}X$

We want  $U(X)$   $\mathbb{P}(\mu \leq U(X)) \geq 1-\alpha$

(3) Common situation:  $X$  = height/weight of a population  $\mu = \mathbb{E}X$

want  $\mathbb{P}(L(X) \leq \mu \leq U(X)) \geq 1-\alpha$

Def.:  $X = (X_1, \dots, X_n) \sim f(x|\theta)$ ;  $\theta \in \Theta \subseteq \mathbb{R}^k$

A subset  $C(X)$  (not involving unknown parameter) of  $\Theta$  is called a  $(1-\alpha)$  level confidence set (or confidence set with  $(1-\alpha)$  level of significance) for  $\theta$  if

$$\underbrace{\mathbb{P}_{\theta}(\theta \in C(X))}_{\text{true coverage probability}} \geq 1-\alpha \quad \forall \theta \in \Theta$$

true coverage probability

$$\Leftrightarrow \inf_{\theta \in \Theta} P_{\theta}(\theta \in C(X)) \geq 1 - \alpha$$

Confidence coefficient of  $C(X)$  is  $\inf_{\theta \in \Theta} P_{\theta}(\theta \in C(X))$   
worst coverage prob.

$$X_1, \dots, X_n \sim N(\mu, 1)$$

$$\text{C.I. for } \mu \quad [\bar{X} - 1, \bar{X} + 1]$$

$$\begin{aligned} & \inf_{\mu} P_{\mu}(\bar{X} - 1 \leq \mu \leq \bar{X} + 1) \\ &= \inf_{\mu} P\left(-\frac{1}{\sqrt{1/4}} \leq \frac{\bar{X} - \mu}{\sqrt{1/4}} \leq \frac{1}{\sqrt{1/4}}\right) \\ &= 2\Phi(2) - 1 \approx 0.95 \end{aligned}$$

Observe :  $\bar{X} = 1.5$     C.I. =  $[0.5, 2.5] \rightarrow$  observed c.i.

Two ways to construct confidence set

- (I) Pivotal Quantity
- (II) Inverting a test

Approach I: Def.:  $X \sim f(x|\theta)$ . A random variable  $Q(X, \theta)$  is called a pivotal quantity (P.Q.) or a pivot for  $\theta$  if the distn of  $Q(X, \theta)$  does not depend on  $\theta$

Ex:  $X_1, \dots, X_n \sim N(\theta, 1)$

A P.Q. for  $\theta$ ,  $Q = \bar{X} - \theta \sim N(0, 1/n)$

$$2) X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$$

$$Q = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$$

So  $Q$  is a P.Q. for  $(\mu, \sigma^2)$ ,  
not for  $\mu$  or  $\sigma$  only.

In general

$$X_1, \dots, X_n \sim f_X(x|\theta) = \frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right) \text{ where } f = \text{known pdf}$$

$$\mu \in \mathbb{R}, \sigma > 0$$

$$1) \text{ Pivot for } \mu: Q_1 = \bar{X} - \mu, X_{(n)} - \mu$$

(if  $\sigma$  known)

$$2) \text{ Pivot for } \sigma: Q_1 = \frac{S}{\sigma}, \frac{X_{(n)} - X_{(1)}}{\sigma}$$

(if  $\mu$  known)

$$3) \text{ Pivot for } \mu: Q_1 = \frac{\bar{X} - \mu}{S} \text{ is a P.Q. for } \mu$$

(when  $\mu, \sigma$  unknown)

$$Q_2 = \frac{S}{\sigma} \text{ is a P.Q. for } \sigma$$

Approach I:

Step 1: Find a P.Q., say  $Q(X, \theta)$  for  $\theta$

Step 2: Find  $a, b$  s.t.  $P_\theta(a \leq Q(X, \theta) \leq b) \geq 1 - \alpha$

Step 3: Define  $C(X) := \{\theta \in \Theta: a \leq Q(X, \theta) \leq b\}$

Claim:  $C(X)$  is a  $(1 - \alpha)$  confidence set for  $\theta$ .

$$\inf_{\theta \in \Theta} P_\theta(\theta \in C(X)) = \inf_{\theta \in \Theta} \underbrace{P_\theta(a \leq Q(X, \theta) \leq b)}_{\text{free of } \theta} \geq 1 - \alpha$$

Ex:  $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$

$$Q = \sqrt{n}(\bar{X} - \mu) \text{ is a P.Q. for } \mu$$

$$\sim N(0, 1)$$



We can choose  $\begin{cases} a = z_{1-\alpha/2} \\ b = z_{\alpha/2} \end{cases}$

$$\begin{aligned} C(X) &= \left\{ \mu : \underbrace{z_{1-\alpha/2}}_{-z_{\alpha/2}} \leq \sqrt{n}(\bar{X} - \mu) \leq z_{\alpha/2} \right\} \\ &= \left[ \bar{X} - \frac{1}{\sqrt{n}} z_{\alpha/2}, \bar{X} + \frac{1}{\sqrt{n}} z_{\alpha/2} \right] \end{aligned}$$